# Chapter 1 · Energy Causes Change

**Teacher Key · Guided Notes**

**Name:** _____ **Homeroom:** ○ 4A ○ 4B ○ 4C ○ 4D

**KEY** *Bold brown = exact answer · Italic gray = sample / acceptable response.*

## BQ Where can we observe evidence of energy causing change?

### My first thinking

*Sample: "A pencil gets shorter and sharper." (Accept any observed change.)*

## V1 VOCABULARY – ENERGY & CHANGE

*Reader p. 1*

energy – the ability to cause **change** .

change – to become **different** .

## N1 ENERGY CAUSES CHANGE

*Reader pp. 1–2*

**Energy has the ability to cause change .**

*When you see a change, you are seeing evidence of energy.*

**ENERGY** → **CHANGE** → **EVIDENCE**

A change is **evidence** that energy is involved.

## N2 ENERGY OCCURS IN DIFFERENT FORMS

*Reader p. 3*

*Write or draw one example that shows a change for each form.*

| FORM | ONE EXAMPLE THAT SHOWS A CHANGE |
|---|---|
| **motion** | *a ball rolls across the floor* |
| **sound** | *a drum makes a noise* |
| **light** | *a flashlight lights up the room* |
| **heat** | *a warm mug warms my hands* |
| **electrical energy** | *a fan spins when plugged in* |

## V2 VOCABULARY – STORED ENERGY

*Reader p. 4*

stored energy – energy that has the ability to cause change at a **later** time.

## N3 OBJECTS CAN STORE ENERGY

*Reader pp. 4–5*

**STORED ENERGY** → **RELEASE** → **CHANGE**

| EXAMPLE | WHAT CHANGE COULD HAPPEN LATER? |
|---|---|
| **battery (in a flashlight)** | *the flashlight lights up* |
| **stretched rubber band** | *it snaps and makes a sound* |
| **raised object (apple)** | *it falls down* |

Objects also have stored energy because of their **position** .

## N4 RECOGNIZING EVIDENCE

*Reader pp. 1–6*

*Draw a change you observed. Show what it looked like BEFORE and AFTER.*

| BEFORE | AFTER |
| --- | --- |
| | |

**What changed?**

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## CHECK YOUR THINKING

### How do we know energy is involved if we cannot see energy itself?

*Start with: "We know because we see …"*

*Sample: "We know because we see a change, like a ball starting to move or a light turning on." (Accept any response connecting a change to evidence of energy.)*

## BQ · NOW I THINK Where can we observe evidence of energy causing change?

*Sample: "We observe evidence of energy causing change everywhere — things move, make sound, light up, heat up, or change shape." (Accept any response naming a change as evidence.)*